

Witness Name: Professor Graham
Medley
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COVID-19 INQUIRY – MODULE 7

WITNESS STATEMENT OF PROFESSOR GRAHAM MEDLEY

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I, **PROFESSOR GRAHAM MEDLEY**, of the London School of Hygiene and Tropical Medicine ('LSHTM') on Keppel Street, London WC1E 7HT, will say as follows:

Section 1: Introduction

- 1.1. I am providing this written statement in response to the Inquiry's Rule 9 request dated 1 November 2024 (**the Rule 9 request**) in relation to Module 7: Test Trace and Isolate.
- 1.2. I previously submitted a response to the Inquiry's Rule 9 Questionnaire on 26 September 2022 (**The Rule 9 Questionnaire Response**) in relation to Module 2: Core UK decision-making and political governance [INQ000056535].
- 1.3. Matters I set out within this statement are within my own knowledge save for where I state otherwise. Where I refer to facts not within my own knowledge, I will provide the source for those facts. The contents of this statement are true, to the best of my knowledge and belief.
- 1.4. This statement addresses my role and involvement in the approach to testing, tracing and isolation ('TTI') adopted during the pandemic in England, Wales, Scotland and Northern Ireland from 1 January 2020 until 28 June 2022. This statement has been prepared based on my personal recollections, and the personal views expressed are my own.

Professional background

- 1.5. I have a BSc in Biology and Computer Science from the University of York (1982) and a PhD from Imperial College London (1989). I have been researching the transmission of dynamics and epidemiology of infectious diseases since 1983. I was a member of the Neglected Tropical Disease modelling consortium until September 2024, where modelling was utilised to inform global and national policy. An example of the type of work we advised on includes the infectious disease visceral leishmaniasis ('VL') which was relevant particularly to the Indian subcontinent.
- 1.6. Between 2017 and 2023 I was chair of the Scientific Pandemic Infections group on Modelling ('SPI-M'), which was the pandemic preparedness committee in the former UK Department of Health (which became the Department of Health and Social Care ('DHSC') in 2018).
- 1.7. Between 2018 and 2023 I was a member of the Statistical Experts Group for the Infected Blood Inquiry. The group provided information and guidance regarding the number of people infected [GM-7/001 – INQ000535653].

- 1.8. Between 2020 and 2022, I co-chaired the Scientific Pandemic Influenza Group on Modelling, Operation sub-group ('**SPI-M-O**') of Scientific Advisory Group for Emergencies ('**SAGE**') which provided evidence to the UK COVID-19 response.
- 1.9. The primary objective of SPI-M-O was to provide evidence from modelling of the transmission dynamics of COVID-19, and explain our understanding, including relative magnitudes, durations and patterns. Modelling has the potential to explore the duration of an epidemic (or wave) and understand the influence of different outcomes and decisions on future patterns.
- 1.10. From November 2020 until March 2022, I was invited to the Wales Technical Advisory Group ('**Wales TAG**') to attend as the co-chair of SPI-M-O to help ensure that the modelling in Wales was consistent with SPI-M-O.
- 1.11. From August 2020 until January 2022 I was invited, along with other SPI-M-O members, to attend meetings with NHS England ('**NHSE**') to discuss the interpretation of modelling for health service planning.
- 1.12. Between March 2020 and July 2021, I attended many meetings with the Joint Biosecurity Centre ('**JBC**') and UK Health Security Agency ('**UKHSA**'), in my position as co-chair of SPI-M-O. My role was to support their development of the production of national and sub-national reproduction numbers and medium-term projections, having been through the experience of developing and publishing these statistics with SPI-M-O. The focus of these meetings was not TTI so I will not discuss in any further detail.
- 1.13. I attended meetings organised by the Academy of Medical Sciences during 2020 and 2021 to discuss the following reports: Preparing for a Challenging Winter 2020/21 [GM-7/002 – INQ000440246] and COVID-19: Preparing for the Future (2021) [GM-7/003 – INQ000064058]. These reports were commissioned by SAGE.

Career history

- 1.14. I have been a professor at London School of Hygiene and Tropical Medicine since 2015. Prior to that, I have held the following appointments:
 - a) Lecturer, Reader and Professor at the University of Warwick from 1997 – 2014;
 - b) Lecturer and Royal Society University Research Fellow at the University of Warwick from 1993 – 1997;

- c) Honorary Lecturer & Royal Society Research Fellow at Imperial College London from 1989 – 1993;
- d) Research Assistant in the Department of Biology at Imperial College London from 1983 – 1989; and
- e) Research Associate at the Department of Fire Safety Engineering at the University of Edinburgh from 1982 – 1983.

Awards and prizes

1.15. I have been awarded the following prizes for my work:

- a) Joint award of the 1990 “Outstanding Statistical Application” from the American Statistical Association for Medley, Billard, Cox & Anderson (1988) in 1990;
- b) Royal Society University Research Fellowship, and extension until 1997 in 1993 in 1989,
- c) OBE for services to the COVID-19 response in 2020;
- d) Weldon Memorial Medal jointly to SPI-M-O, and
- e) Royal Society Gabor medal for leading an interdisciplinary team of biologists, clinicians, mathematicians and statisticians who provided SAGE with epidemiological modelling expertise concerning the COVID-19 pandemic.

Professional expertise

1.16. I have worked on a wide variety of infections using a wide variety of techniques and approaches. I have provided policy advice to HMG, most notably on HIV/AIDS and vCJD. I have over 220 peer-reviewed scientific publications on the transmission dynamics of infectious diseases. My first publication was in 1983.

Major publications relevant to TTI

1.17. A complete list of my relevant publications can be found in Annex A, however the publications I consider directly relevant to TTI are set out below:

- a) **‘Public perceptions and interactions with UK COVID-19 Test, Trace and Isolate policies, and implications for pandemic infectious disease modelling’**, Guy C. Marshall, Miguel E. P., Martyn Fyles, Thomas House, Emma L. Davis, Li Pi, Graham F. Medley, Billy J. Quilty, Louise Dyson, Lucy Yardley, Elizabeth Fearon, 6 September 2022 [GM-7/004 – INQ000535675].

- b) **‘Contact tracing is an imperfect tool for controlling COVID-19 transmission and relies on population adherence 2021’**, Emma L. Davis, Tim C. D. Lucas, Anna Borlase, Timothy M. Pollington, Sam Abbott, Diepreye Ayabina, Thomas Crellen, Joel Hellewell, Li Pi, CMMID COVID-19 Working Group, Graham F. Medley, T. Déirdre Hollingsworth & Petra Klepac, 13 September 2021[GM-7/005 – INQ000535676].
- c) **‘Implication of backward contact tracing in the presence of overdispersed transmission in COVID-19 outbreaks’**, Akira Endo, Centre for the Mathematical Modelling of Infectious Diseases COVID-19 Working Group, Quentin J Leclerc, Gwenan M Knight, Graham F Medley, Katherine E Atkins, Sebastian Funk, Adam J Kucharski, 31 March 2021 [GM-7/006 – INQ000535677].

Articles

- 1.18. I have written a commentary titled ‘A consensus of evidence: the role of SPI-M-O in the UK COVID-19 response’ for the journal Advances in Biological Regulation special issue [GM-7/007 – INQ000213281].

Section 2: Involvement in SAGE, SPI-M, SPI-M-O, NHS, JBC and UKHSA and Research

SPI-M

- 2.1. As set out at paragraph 1.6, between 2017 and 2023 I was academic chair of SPI-M. The role of SPI-M was to provide modelling evidence to support the DHSC in preparation of a pandemic. Initially this was based on influenza however this changed focus to infectious epidemic during 2019. The 2018 report focuses on the initial stages of an epidemic, so does not address TTI, nor was TTI part of the discussion of preparedness plans [GM-7/008 – INQ000212124].

SPI-M-O

- 2.2. In January 2020 SAGE established the subgroup SPI-M-O. I was invited to co-chair due to my experience in infectious diseases and my position as co-chair of SPI-M. I was co-chair until February 2022, when I returned to co-chair SPI-M. SPI-M-O met over 100 times from January 2020 until 23 March 2022 and I attended those meetings as academic co-chair. The initial policy co-chair was Professor Jonathan Van-Tam (then Deputy Chief Medical Officer), and he was replaced by Professor

Dame Angela McLean in March 2020. As per the January 2020 Terms of Reference for SPI-M-O [GM-7/009 – INQ000535680], my responsibilities as co-chair included:

- a)* Discussing the agenda and advising on how to present matters to get the best engagement;
- b)* Highlighting and recommending areas that I thought modelling could assist with;
- c)* Ensuring that the secretariat had sufficient information and understanding to draft the consensus statement;
- d)* Commenting on and approving the consensus statement to ensure that it was a good and fair reflection of members' discussion. Particular attention was given to the uncertainty statements;
- e)* Addressing changes to consensus statements after SAGE meetings;
- f)* Attending meetings to represent SPI-M-O and to some extent 'modelling'; and
- g)* Commenting on and approving the minutes of SPI-M-O meetings.

SAGE

2.3. I was a participant of SAGE from January 2020 attending in my capacity as co-chair of SPI-M-O. I attended 97 meetings [GM-7/010 – INQ000535654]. I was regularly invited to attend further meetings until SAGE stood down in February 2022.

2.4. I also attended regular meetings from other subgroups which included:

- a)* **Enduring prevalence** – Within this group, discussions were based on the various areas in the country which had high prevalence, and when it dropped, it came back up quicker again. I attended along with Professor Julia Gog who was SPI-M-O member at Cambridge. Our main role was to bring in the quantitative understanding of the data. As a result, we had the testing data (not tracing) at a fairly fine geographical scale, and we were able to bring along SPI-M-O modelling and data analysis to address some of the issues of enduring prevalence as much as we could. We were restricted in provided further modelling information due to the lack of complete and accurate data from TTI;
- b)* **Care homes** - This sub-group commenced in April 2020 and the main focus was to understand what was happening in care homes. The focus then

changed to understanding what to do and testing formed part of that. I was only involved initially until Professor Ian Hall (SPI-M-O member, Manchester) became more involved and took over as co-chair;

- c) **Schools/children** - Within this group, the focus was understanding the spread of COVID-19 and the use of Lateral Flow Testing ('LFT'). I was involved in initial discussions before having to move away as the epidemic developed;
- d) **Higher and further education** - As above, the focus of this group was understanding the spread of COVID-19;
- e) **Nosocomial Transmission** - This group focused on the spread of COVID-19 within hospitals. I was initially involved as co-chair before passing this on to individuals who were working within hospitals, who were better placed to advise on this, especially Dr Julie Robotham (UKHSA). The group were reporting separately and directly into SAGE;
- f) **Mass testing** - This sub-group existed for a very short period of time; and
- g) **Vaccinations** - It was important for information to be shared, therefore this group existed to provide updates and information on vaccines rather than providing analysis or advice. Vaccinations were being dealt with a different set of scientific advisers and were not going through SAGE.

- 2.5. None of these groups specifically focused on TTI although it will have been discussed occasionally. Better data from TTI would have greatly informed the discussions on Enduring Prevalence, providing, for example, information on connectedness of different areas (such as the distance between addresses of index cases and their contacts) and importance of settings such as workplaces for transmission (such as the proportion of contacts made at work). TTI would have been useful for understanding the links between the community, care homes, schools and hospitals.
- 2.6. I also attended regular meetings between the chairs of all SAGE subgroups to discuss ongoing work to keep information flowing between us all.
- 2.7. It is important to note that SAGE and SPI-M-O did not make decisions or policies. SPI-M-O provided evidence from modelling rather than advice. I was very conscious that my role was to present the evidence from SPI-M-O rather than develop the

evidence into advice. During SAGE meetings I highlighted when I was speaking as an individual rather than representing SPI-M-O.

- 2.8. During the pandemic, the Government received advice from many other experts including PHE/UKHSA, NHS Test and Trace, DHSC, the Cabinet Office communications team, advertising and market research agencies and went on to make decisions based on a wide range of factors.

NHS

- 2.9. From August 2020 I was invited, along with other SPI-M-O members, to attend meetings with NHSE. These were largely informal meetings primarily for NHSE to be able to ask clarifying questions directly to SPI-M-O members, and for members to raise issues and get a better understanding of health-care challenges.
- 2.10. I attended one meeting with NHSX on 4 April 2020 to discuss contact tracing. Minutes were kept by NHSX.

Wales TAG

- 2.11. In my capacity as co-chair of SPI-M-O, I offered advice and help to Wales TAG in developing modelling capacity and interpretation of evidence. I read and commented on draft documents. I kept the SPI-M-O secretariat and others informed of developments in Wales that were of interest to the remainder of the UK and ensured that the modelling in Wales was consistent with SPI-M-O.

JBC & UKHSA

- 2.12. Between March 2020 and July 2021, I attended JBC and UKHSA meetings, principally in my position as co-chair of SPI-M-O having been through the experience of developing and publishing estimates of reproduction numbers and medium-term projections. The reproduction number (*'R'*) is the average number of secondary infections produced by a single infected person.
- 2.13. My contribution was to advise and support development of the infrastructure for providing and publishing estimates of reproduction numbers.

Research

- 2.14. During the COVID-19 pandemic I mentored Dr Elizabeth Fearon's Medical Research Council research fellowship at LSHTM. Dr Fearon successfully obtained a research grant, which started on 1 September 2020, titled *'An analytical framework for Test, Trace and Isolate in the UK: optimising and targeting deployment alongside other*

measures'. I interacted with Dr Fearon throughout, and, although most of the work was done jointly with researchers in Manchester, I attended discussions, and co-wrote papers with her. Dr Fearon and colleagues worked more closely with TTI than I did.

- 2.15. At the start of the epidemic and until January 2021 I was director of the Centre for Mathematical Modelling of Infectious Disease ('**CMMID**') at LSHTM. CMMID set up a co-operative publication system using the group author "CMMID COVID-19 Working Group" on all publications to acknowledge the background work that contributed to all the output from CMMID. I was a member of this group authorship. I commented on papers as they were written.
- 2.16. At the start of the epidemic until February 2020 I was partly seconded from LSHTM to University of Oxford to support the management of the Neglected Tropical Disease modelling consortium. Consequently, I was involved with research coming from this group.

Section 3: Testing Technologies

- 3.1. I am asked to set out any relevant evidence in relation to testing technologies deployed during the pandemic. This includes the development and comparison of assays, testing systems and strategies for the pandemic, the role of scientific modelling in the TTI systems, and testing systems and strategies for future pandemics.
- 3.2. My role did not include any involvement with assays so I am unable to comment on the development of technologies involved in this area.

Testing systems and strategies during the pandemic

- 3.3. From my perspective during the COVID-19 epidemic, the key considerations of the TTI system were:
 - a) surveillance, and
 - b) control of transmission.
- 3.4. I have discussed each of these in further detail below, with examples of the advice given during the pandemic.

Surveillance

- 3.5. Surveillance testing provides data that can be used to estimate the incidence and prevalence of infection within different segments of the population. The two principal

data streams introduced in the second quarter of 2020 were the Office of National Statistics (**ONS**) Covid Infection Survey and the 'Real-time Assessment of Community Transmission' (**REACT**), both of which were introduced by the scientific community, although I had no direct role in the design or operation of either. It was important to have two different data streams so that they confirmed each other, given they both rely on people being tested at random, with no prior information on their infection status. These testing surveillance data were used heavily by the SPI-M-O members and provided considerable information which was used in the production of evidence through SAGE to support government decision-making. In future epidemics such systems should be introduced before the first case, and having two independent mechanisms provides evaluation and ability to cross-check data.

- 3.6. Areas of low prevalence provided little information to assess their status through health-care data streams (i.e. hospital admissions). Directing testing to low prevalence areas would provide a bigger gain of information relevant for policy and decision-making, and in particular whether the incidence is increasing or not. Dynamically an epidemic is in the same situation if the number of cases doubles from 10 to 20 as it is from 10,000 to 20,000, but the former is much harder to estimate. This might have played into policy understanding during the summer 2020 when prevalence was starting to increase but there was not sufficient data to reduce uncertainty. With hindsight, TTI might well have provided additional community data stream, and is an additional argument for directing testing resources to areas of lower prevalence (see paragraph 3.10 below).

Control of transmission

- 3.7. Control of transmission through TTI relies on identifying infected people so that they can reduce transmission. The TTI system developed during COVID-19, including the NHS COVID-19 app, was an intervention measure. The testing data (number of positive and total tests) were used by SPI-M-O members as data streams to inform models. They were a very useful data source, given their population coverage, but would have been more useful with fairly small changes to the way the data were collected and presented. For example, linkage with hospital admissions, whether positive cases had been asked to isolate the previous week and the reason for testing would have been very informative. The people in control of the TTI system are best placed to understand the biases and temporal changes in the data, but this understanding was not made clear.

- 3.8. Reaching the maximum testing capacity is almost inevitable during an epidemic. Constraints of any testing system need to be factored in at the planning stage. In particular it is important that tests are reported swiftly as the delay in testing is a major constraint on the impact of testing. Testing capacity was reached at several points during the epidemic. The SPI-M-O consensus statement on 16 September 2020 taken to SAGE 57 stated *“Recent operational issues with NHS Test and Trace have exacerbated SPI-M-O’s uncertainty in their estimates of R , growth rates, doubling time and incidence. The service currently appears to be at capacity, which is resulting in two important complications. The demand for symptomatic testing is not being met, so pillar 2 cases are subject to a time-dependent constraint and observed trends are not a simple reflection of the level of positive cases. Just as important for interpretation, delays in testing have changed and those changes in delays may be different for positive and negative results.”* [GM-7/011 – INQ000223534].
- 3.9. During an epidemic it is almost inevitable testing capacity will be reached, and the design of the TTI system should factor this in. The priority is to maintain the effectiveness of those tests that can be done, ensuring that the delays in testing do not lengthen as the testing volume increases or reaches its maximum. Inevitably there is also a question of allocation of the testing resources, and some mechanism for prioritisation of which tests are to be performed.
- 3.10. Between August 2020 and October 2020 SPI-M-O were asked about the impact of resource limitations and the optimum allocation of available testing capacity. The principal outcome of the research and modelling from a strategic viewpoint was that TTI can reduce the R for COVID-19, by at best one third. The R number has a threshold at one. Above that the epidemic is increasing exponentially, and if the R is below one the epidemic is decreasing. If the aim of TTI is to reduce the total number of infections, or minimise the number of places with high prevalence, then TTI should be focussed on those areas where the prevalence is low and the R is less than three so that TTI can bring R below one. These results were reported in the SPI-M-O consensus statement of 21 October 2020 (paragraphs 21-25) sent to SAGE 62 [GM-7/012 – INQ000213277].

Contact Tracing App

- 3.11. On 4 April 2020 I attended an NHSX meeting to discuss the development of an app to support contact tracing. I was not closely involved in the development of the app, which went on to become the NHS track and trace app. I believe Professor

Christophe Fraser (Big Data Institute, University of Oxford) was directly involved in the development of the app, but I am not aware of any SPI-M-O members being closely involved.

- 3.12. Personally, I believe that the use of a contact tracing app was critically useful as an intervention. It could have been more informative for modelling, but it was neither intended nor designed as a research and surveillance tool. An issue that was raised was whether the definition of a contact should be determined by person or place. My understanding is that data restrictions would not allow both. The app was able to record a contact or where contact had occurred, but not both. The app used contact (i.e. the people who had been in close enough contact), but to my knowledge there was no evaluation of the impact of using place (i.e. declaring all people who were in the same setting at the same time as contacts). Using place would very likely increase the numbers of contact per case but might result in more effective intervention.
- 3.13. In my opinion, the contact tracing app and TTI had a big influence on the transmission of COVID-19 in the period July to November 2021. During this period, the incidence of infection (as measured by the rate of hospitalisations) was broadly constant. This situation, where each infection gives rise to one other infection on average, is unusual. Undoubtedly vaccine and infection-derived immunity were important determinants, but it was also clear to me, through informal monitoring of friends and family, that people were using both the formal interventions and informal contact monitoring, to understand and manage their risks.
- 3.14. One of the biggest impacts of the contact tracing app was during July 2021 – the so-called ‘pingdemic’. This is when contacts who were identified as being in contact with someone who tested positive for COVID-19 were required to isolate. This attracted considerable public interest. The impact on certain sectors of the economy was significant. The requirement for contacts to isolate will have had two important consequences. Firstly, there will have been an impact on transmission from contacts who were infected, but not knowing the proportion of infections that were notified prevents direct estimation of the size of this impact as explained in paragraph 3.7 [GM-7/013 – INQ000561521]. Secondly, contacts who were not infected will have been removed from circulation and therefore unable to be infected during their isolation period which is referred to as ‘susceptible thinning’. The effect of this will have been enhanced because those notified will have been in networks within which COVID-19 was circulating. Overall, it is my belief that contact tracing would have

been a far more efficient intervention than population wide restrictions (lockdown) in that it was much better targeted.

- 3.15. The ultimate aim in an epidemic is for people to effectively manage their own risks of infection, rather than government managing those risks through blanket measures such as 'lockdown'. The contact tracing app provided the most immediate information to people about their risks and tied in with personal, informal communications. Although the app and TTI system likely had significant impact on transmission during July-November 2021, I believe that there was a missed opportunity to develop the app for use in future epidemics. This includes development of digital management of transmission, such as the risk of personal contact vs. the risk of place of contact, duration of exposure and many other parameters. It also includes using digital approaches to monitoring contact rates, and providing users with risk-related information, such as the number of contacts in the previous day. The data gathered during the study that underpins the publication shows how informative such approaches can be [GM-7/013 – INQ000561521].

Role of scientific modelling in the development, implementation, and evaluation of TTI systems

- 3.16. As previously explained, SPI-M-O provided evidence rather than advice. There was limited capacity to understand the value and potential of modelling within government, at least until January 2021. Perhaps the biggest reason for the change from January 2021 was a more constructive dialogue between SPI-M-O and Cabinet Office about the aims and intentions of policy as explained in my Module 2 statement at paragraph 8.46 and 8.54-8.59 [INQ000260643].
- 3.17. Early in the epidemic, there was considerable scientific interest in understanding how TTI (called 'contact tracing' generically in the scientific literature) might interact with other interventions. For example, the following paper was presented to SPI-M-O on 20 April 2020 titled '*Effectiveness of isolation, testing, contact tracing, and physical distancing on reducing transmission of SARS-CoV-2 in different settings: A mathematical modelling study in June 2020*' [GM-7/014 – INQ000255418] alongside a similar paper from Imperial College. This resulted in a SPI-M-O consensus statement 20 April 2020 paragraphs 7-14 taken to SAGE 27, which outlined the principal requirements of TTI [GM-7/015 - INQ000213298].

Testing strategies

- 3.18. Given that the requirement to isolate for 14 days was itself harmful, costly and difficult, there was considerable interest in using testing to determine when individuals could be released based on negative test results. The results of one modelling report were discussed at SPI-M-O 17 June 2020 and reported in SPI-M-O consensus 17 June 2020 (taken to SAGE 42) [GM-7/016 – INQ000253878]. Serial testing of cases did impact on testing capacity, although the arrival of LFT largely resolved this. There was some work in November 2020 to evaluate the use of daily LFT testing as an alternative to self-isolation reported in SPI-M-O consensus 11 November 2020 sent SAGE 67 paragraphs 16-26 [GM-7/017 – INQ000253895].

Mass screening

- 3.19. There was considerable policy interest in the use of mass screening (testing a population regardless of symptoms or apparent risk of infection). This resulted in a working group of SAGE to which I contributed [GM-7/018 – INQ000213316]. This document explained that although this approach is different from TTI (then called NHSTT) the two systems should interact in terms of information flow, and the optimum use of testing capacity.
- 3.20. The Liverpool evaluation of mass testing demonstrated it was potentially useful, and in particular demonstrated the value of LFT in large-scale testing. The principal challenge (as for any voluntary intervention) was willingness to test and the consequent interaction with socio-economic inequalities (especially when self-isolation resulted in loss of earnings) [GM-7/019 – INQ000223452].

Section 4: Lessons that can be learned and recommendations for further changes relevant to Test Trace and Isolate.

- 4.1. Mass testing and screening have the potential to be important parts of an overall strategy but are unlikely to be effective as single measures and needs to be carefully implemented and integrated with more targeted (risk-based) approaches.
- 4.2. Testing provides information for targeted interventions, which could make the difference between locking down a whole population or a section of people. In any intervention that is targeted based on testing, the uptake of tests and compliance with the interventions for people who test positive are always the major factors in determining the impact and cost-effectiveness. If mass testing or screening, or more focussed TTI are to be a potential intervention in future epidemics, then there needs to be full consideration prior to the next epidemic of the barriers to uptake, compliance and adherence.

- 4.3. There was discussion about the evaluation and effectiveness of different TTI strategies in the 5 August 2020 SPI-M-O consensus statement (paragraphs 25-30) which went to SAGE 50 [GM-7/020 – INQ000535665]. The issues raised in this consensus statement should, ideally, be raised during design of the TTI system. In particular, SPI-M-O was focussed on improving the data for modelling of the epidemic, and highlighted the data collection that would support this wider objective. Crucially, it became clear that evaluation of the impact of TTI had not been considered at the design stage of TTI during summer 2020 which will have hampered the improvement of the system.
- 4.4. At the latter stages of the epidemic in May 2021 I, along with others, met with a quantitative analyst working at DHSC. He was attempting to evaluate the impact of TTI on transmission dynamics and seeking support. He was, if I remember correctly, using a large spreadsheet, and overall gave the impression that he was a relatively junior analyst working alone on a huge question. The problem he was addressing did not appear to be well posed. I do not believe that we had a quantitative understanding of the impact of TTI during the epidemic, which influenced SPI-M-O's ability to provide evidence. For example, the Imperial College submission regarding the final road map step submitted on 7 July 2021 noted that “...*the impact of Test Trace Isolate (TTI), mask wearing, hand hygiene, and COVID security on R [was] difficult to quantify...*” [GM-7/021 – INQ000262611].
- 4.5. I was a member of the Data Science Advisory Board, part of JBC (which later migrated into UKHSA) from September 2020 until October 2022. In July 2021 the advisory board received a report and presentation from the same analyst of the more developed model. This is the most complete evaluation of TTI that I have seen. It only included data up to April 2021, and it would be very interesting to see this updated to include the period to end of 2021 [GM-7/022 – INQ000535667].
- 4.6. During the summer of 2020 it was raised by multiple SPI-M-O members that data-driven evaluation of TTI was not apparent. The concern was that if evaluation of the impact of TTI was not being measured, then the system could not be improved. As an example, one of the pieces of information that would have been useful would have been the proportion of people entering hospital who had been traced as being in contact with someone who tested positive for COVID-19, as this would have enabled a better understanding of how well TTI was ‘penetrating into’ the epidemic [GM-7/023 – INQ000216143]. I do not believe that these data were routinely collected. This email also highlights that on SAGE, a distinction was also drawn

between 'operational' and 'scientific' issues. Whilst the running of TTI was clearly outside of SAGE's remit, its impact on transmission dynamics was very much in SPI-M-O's remit. Ideally there would have been better links between the operational side of TTI and the scientific understanding that could have been generated.

- 4.7. Throughout the summer of 2020 SPI-M-O consistently requested access to the data generated by TTI, and in particular access to 'line lists' rather than averaged data. The published data was aggregated across all individuals, whereas line lists provide information for each individual. For example, the aggregated data might report the average delay between sample and test result, and the average age, whereas unaggregated data allows the investigation of the relationship between delay and age. SPI-M-O members Professors Christopher Jewell (Lancaster) and Matthew Keeling (Warwick) wrote a letter to the SPI-M-O secretariat on 7 June 2020 after discussing with SPI-M-O members what was required. The intention was that the letter would be sent within DHSC laying out data requirements. I am not able to confirm if the letter was sent or if there was a response. In September 2020 there were various discussions and meetings throughout the summer requesting access to the data, involving, amongst others, the Chief Medical Officer and Ben Warner (former Special Adviser) [GM-7/024 – INQ000535681]. There were several issues within the data that I was aware of, the most important being the failure to record the reason for testing. This would have enabled an improved interpretation of the data.
- 4.8. To the best of my knowledge, the requested TTI data was never made available to the SPI-M-O members. I believe that not allowing SPI-M-O access affected members accurate assessment of the impact of TTI on the epidemic. With this data we could likely have evaluated the population-level performance of TTI and provide feedback suggestions on how to improve its effectiveness.
- 4.9. The data systems in Wales (SAIL) and Scotland were superior in my opinion, as they allowed the linkage of individual data across different platforms. In future epidemics, I hope that there will be better data systems that allow richer datasets to be generated and appropriately shared. TTI is an intervention rather than a data-gathering system, but being able to determine, in near real-time, the impact of the intervention is critical to ensuring its effectiveness and cost-effectiveness.

Section 5: Unequal Impacts

- 5.1. I have been asked to give evidence on how inequalities and unequal impacts were considered and discussed in the decision-making process. From my recollection, in

terms of modelling, SPI-M-O were never explicitly asked to consider how interventions would impact different socio-economic strata, nor how those strata would influence the interventions. It was also a major failing of infectious disease modelling that the models did not have socioeconomics as a structural dimension. This was highlighted in my Module 2 statement [INQ000260643] at paragraph 8.18-8.19 and 12.27.

- 5.2. As explained above, the use of a contact tracing app was critically useful as an intervention, however, there is clearly a question of inequality of access to technology that needs to be addressed. The ability, or degree of difficulty associated with uptake, compliance and adherence to any TTI system relates especially to economic inequalities which differentially enable people to react to unknown risks. The SPI-M-O consensus 10 March 2021 paragraphs 13-23 (sent to SAGE 83) discusses the evidence of the role of self-testing and its interaction with uptake and adherence [GM-7/025 – INQ000535670].
- 5.3. It was known prior to the epidemic that inequalities are critical in determining both the impact of infectious disease, and the effectiveness of interventions [GM-7/026 – INQ000213146]). Infectious disease also plays an important role in creating and maintaining inequalities. It became clear during the epidemic that much of the fine scale spatial patterning of infection was being driven by socio-economic inequalities. By June 2021 data quality had greatly improved (compared to early 2020) which meant that the analysis of the epidemic of the delta variant could clearly be related to socio-economic inequalities as presented in SPI-M-O consensus statement 2 June 2021 paragraphs 18-22 (taken to SAGE 91) [GM-7/027 – INQ000223896]. This greater granularity of data meant that the analysis of transmission was more accurate, i.e. results can be biased if there is a significant determinant of an epidemic (such as aspects of socio-economic status) which is not accounted for in the analysis.
- 5.4. The analysis of the Enduring Prevalence sub-group of SAGE, and the spatial sub-group of SPI-M-O increasingly included socio-economic variables such as the Index of Multiple Deprivation (IMD). The SPI-M-O consensus statement 24 November 2021 paragraphs 8-12 (sent to SAGE 97) demonstrates the greater understanding, but also highlights that data were required to get a better understanding of the role of inequalities in determining the patterns of transmission and their impact [GM-7/028 – INQ000535673].

- 5.5. Modelling of inequalities requires an understanding of the processes that link the socio-economic status to different behaviours. If models are to improve, then more insight is required into the determinants of behaviour. TTI systems cannot, in my opinion, be successfully designed on a white board, but need to be co-developed with people who will seek to incorporate TTI into their lives, this is evidenced in the following paper 'Public perceptions and interactions with UK COVID-19 Test, Trace and Isolate policies, and implications for pandemic infectious disease modelling', Guy C. Marshall, Miguel E. P., Martyn Fyles, Thomas House, Emma L. Davis, Li Pi, Graham F. Medley, Billy J. Quilty, Louise Dyson, Lucy Yardley, Elizabeth Fearon, 6 September 2022 [GM-7/004 – INQ000535675].
- 5.6. The COVID-19 epidemic inspired great progress in data availability and analytical approaches to including socio-economic inequalities as determinants of patterns and impacts. This needs to continue if future epidemics are not going to be driven by inequalities and to further increase inequalities. In my opinion, a key driver for this would be for government policymakers to make it clearer the extent to which they are prepared to mitigate differential impact by socio-economic status. At present, interventions are evaluated and analysed based on their total or mean impact, but understanding how governments value and wish to use the distributional outcomes of interventions would provide important impetus to develop data systems and analytical tools.

Statement of Truth

I believe that the facts stated in this witness statement are true. I understand that proceedings may be brought against anyone who makes, or causes to be made, a false statement in a document verified by a statement of truth without an honest belief of its truth.

Signed:

Personal Data

Dated: 6 May 2025

Annex A: Major publications

1. 'When an emerging disease becomes endemic', Graham F. Medley, Anna Vassall, 14 July 2017.
2. 'Public perceptions and interactions with UK COVID-19 Test, Trace and Isolate policies, and implications for pandemic infectious disease modelling [version 1; peer review: awaiting peer review]. F1000Research 2022, 11:1005, Guy C. Marshall, Miguel E. P., Martyn Fyles, Thomas House, Emma L. Davis, Li Pi, Graham F. Medley, Billy J. Quilty, Louise Dyson, Lucy Yardley, Elizabeth Fearon, 6 September 2022.
3. 'Contact tracing is an imperfect tool for controlling COVID-19 transmission and relies on population adherence', Emma L. Davis, Tim C. D. Lucas, Anna Borlase, Timothy M. Pollington, Sam Abbott, Diepreye Ayabina, Thomas Crellen, Joel Hellewell, Li Pi, CMMID COVID-19 Working Group, Graham F. Medley, T. Déirdre Hollingsworth & Petra Klepac, 13 September 2021.
4. 'Estimating the effectiveness of routine asymptomatic PCR testing at different frequencies for the detection of SARS-CoV-2 infections', Joel Hellewell, Timothy W. Russell, The SAFER Investigators and Field Study Team, The Crick COVID-19 Consortium, CMMID COVID-19 working group, Rupert Beale, Gavin Kelly, Catherine Houlihan, Eleni Nastouli & Adam J. Kucharski, 27 April 2021.
5. 'Diversity of symptom phenotypes in SARS-CoV-2 community infections observed in multiple large datasets', Martyn Fyles, Karina-Doris Vihta, Carole H Sudre, Harry Long, Rajenki Das, Caroline Jay, Tom Wingfield, Fergus Cumming, William Green, Pantelis Hadjipantelis, Joni Kirk, Claire J Steves, Sebastien Ourselin, Graham F Medley, Elizabeth Fearon & Thomas House, 7 December 2023.
6. 'The local burden of disease during the first wave of the COVID-19 epidemic in England: estimation using different data sources from changing surveillance practices', Emily S. Nightingale, Sam Abbott, Timothy W. Russell, CMMID Covid-19 Working Group, Rachel Lowe, Graham F. Medley & Oliver J. Brady , 11 April 2022.
7. 'Considering equity in priority setting using transmission models: Recommendations and data needs', M. Quaife, GF Medley, M. Jit, T. Drake, M. Asaria, P. van Baal, R. Baltussen, L. Bollinger, F. Bozzani, O. Brady, H. Broekhuizen, K. Chalkidou, Y.-L. Chi, DW Dowdy, S. Griffin, H. Haghparsat-Bidgoli, T. Hallett, K. Hauck, TD Hollingsworth, CF McQuaid, NA Menzies, MW Merritt, A. Mirelman, A. Morton, FJ Ruiz, M. Siapka,

J. Skordis, F. Tediosi, P. Walker, RG White, P. Winskill, A. Vassall, GB Gomez, 4 November 2022.

8. 'Implication of backward contact tracing in the presence of overdispersed transmission in COVID-19 outbreaks', Akira Endo, Centre for the Mathematical Modelling of Infectious Diseases COVID-19 Working Group , Quentin J. Leclerc, Gwenan M. Knight, Graham F. Medley, Katherine E. Atkins, Sebastian Funk, Adam J. Kucharski, 31 March 2021.
9. 'Using a household-structured branching process to analyse contact tracing in the SARS-CoV-2 pandemic', Fyles Martyn, Fearon Elizabeth, Overton Christopher, University of Manchester COVID-19 Modelling Group, Wingfield Tom, Medley Graham F., Hall Ian, Pellis Lorenzo and House Thomas, 31 May 2021.
10. 'Engagement and adherence trade-offs for SARS-CoV-2 contact tracing', Lucas Tim C. D., Davis Emma L., Ayabina Diepreye, Berlese Anna, Crellen Thomas, Pi Li, Medley Graham F., Yardley Lucy, Klepac Petra, Gog Julia and Déirdre Hollingsworth T, 31 May 2021.
11. 'Optimizing time-limited non-pharmaceutical interventions for COVID-19 outbreak control', Morgan Alex L. K., Woolhouse Mark E. J., Medley Graham F. and van Bunnik Bram, 31 May 2021.

Articles

12. 'A consensus of evidence: the role of SPI-M-O in the UK COVID-19 response', Graham F Medley, 11 November 2022.

Opinion pieces/Media

13. Has England reached a peak in Covid infections? Graham Medley, The Guardian 26 July 2021.
14. Is England's Covid roadmap the right way out of lockdown? The experts' view Devi Sridhar, Jennifer Dowd, Graham Medley and Stephen Reicher, The Guardian 22 February 2021.
15. Scientists shouldn't have the final word on Covid-19 plans', Graham Medley, The Times 1 October 2020.

Interviews

16. BBC Newsnight interview, 13 March 2020.

17. 'Boris Johnson's coronavirus adviser calls for a way out of lockdown' The Times interview, 4 April 2020.
18. 'Coronavirus: Pubs 'may need to shut' to allow schools to reopen' BBC Radio 4 interview, 1 August 2020.

Book

19. Coronavirus and Covid: A book for children about the pandemic, 15 April 2021.

Annex B: Exhibit Schedule

Please see attached.